Experiment No: 5 **Date:**

Bluetooth-Based Door Lock System

Aim: Write program for Bluetooth-Based Door Lock System

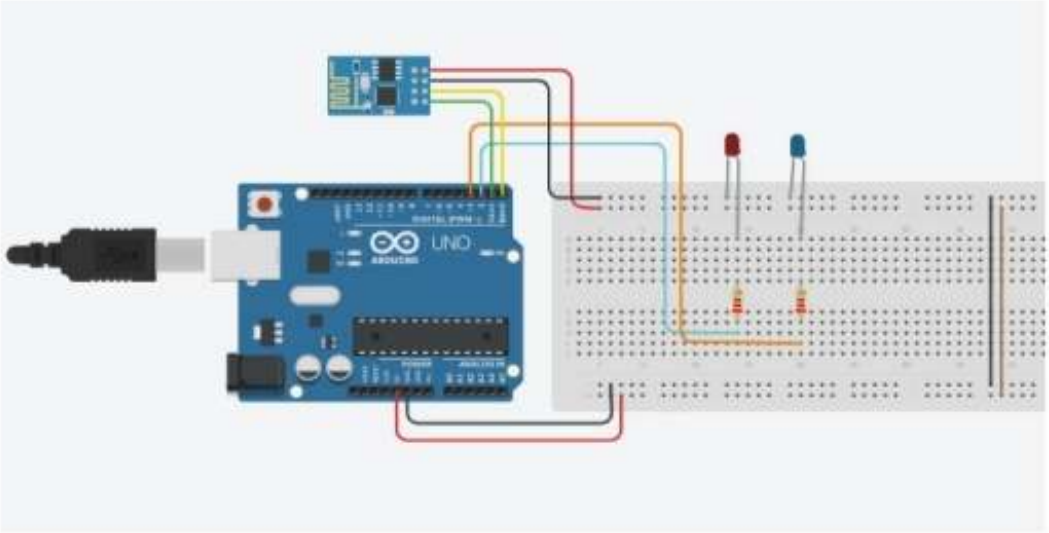
Components Required:

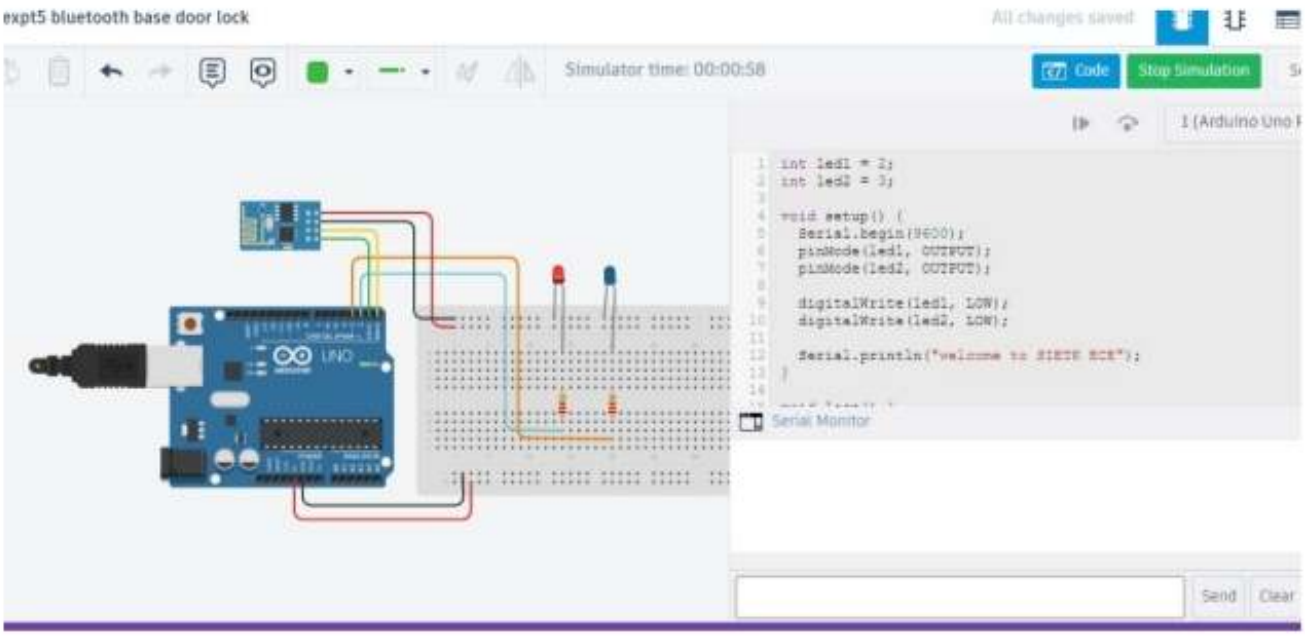
- Arduino Nano
- Bluetooth Module
- Two Leds
- Two Resistors

Theory:

To create a Bluetooth-based door lock system in Tinkercad, need to simulate an Arduino, a Bluetooth module (like HC-05), led to indicate the lock, You will arrange these components in Tinkercad, wire them together, add the appropriate code to your Arduino, and then simulate the system to test its functionality, which involves connecting a phone via Bluetooth and sending commands to unlock the door.

Circuit Diagram:





```
//program for  Bluetooth based door lock system
int led1 = 2;
int led2 = 3;
void setup() {
  Serial.begin(9600);
  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);
  digitalWrite(led1, LOW);
  digitalWrite(led2, LOW);
   Serial.println("welcome to SIETK ECE");
}
void loop() {
  if (Serial.available() > 0) {
    String entrada = Serial.readString();
    entrada.trim();
    if (entrada == "open door1") {
      digitalWrite(led1, HIGH);
      Serial.println("door1 Lock is opened");
    }
    else if (entrada == "Close door1 lock") {
      digitalWrite(led1, LOW);
      Serial.println("Door1 lock is closed");
    }
    else if (entrada == "open door2") {
      digitalWrite(led2, HIGH);
      Serial.println("door2 Lock is opened");
    }
    else if (entrada == "Close door2 lock") {
      digitalWrite(led2, LOW);
      Serial.println("Door2 lock is closedk");
    }
    else {
      Serial.println("both doors are closed");
    } }}
```

Procedure:

1. Open tinker cad
2. Setup the circuit as per circuit diagram
3. Connect the components as shown in the connection diagram.
4. Connect power supply
5. Verify the output

Result:

Experiment No: 11 **Date:**

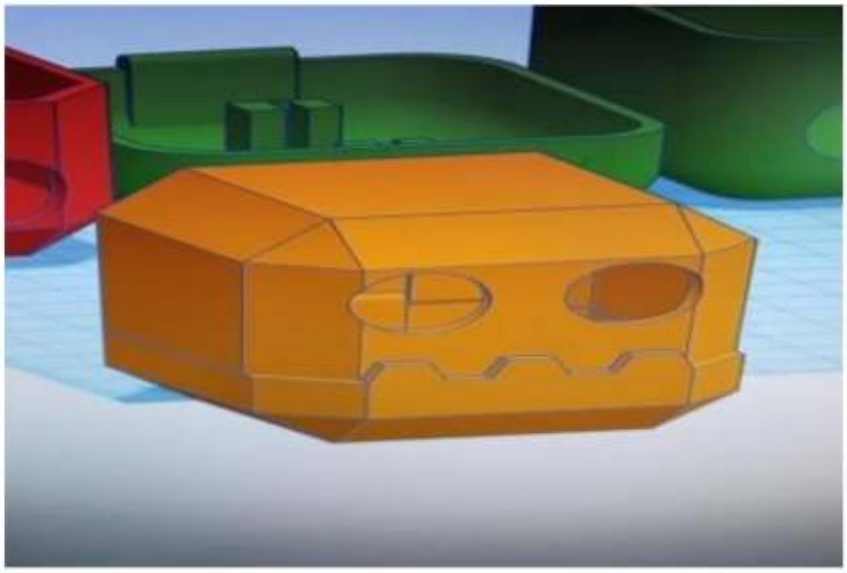
Design a 3D Walking Robot

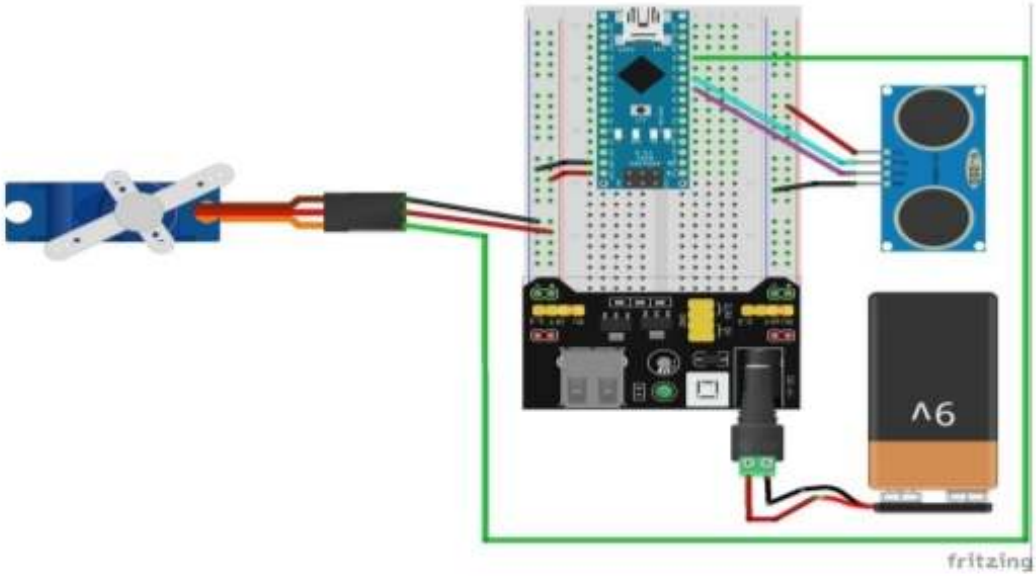
Aim: To Design a 3D print Walking Robot using Tinkercad and Arduino

Components Required:

- Protoboard
- Power supply (I prefer a Protoboard power supply, but you can opt for a battery connected to the Arduino or power the Arduino through the USB cable)
- Arduino Nano
- SG90 Servo
- HC-SR04 ultrasonic distance sensor
- Cables
- 3D printed chassis (Alternatively, you can get creative and use cardboard along with some tape!)

Circuit Diagram:





```
//Program for 3D Robo
#include <Servo.h>// I use the Servo.h library to control servo motors.
// There's no need to install this library; it comes pre-installed in the Arduino programming environmen
#define trigPin 7 // the TRIG pin of the distance sensor is connected to D7
#define echoPin 6 // the ECHO pin of the distance sensor is connected to D6
Servo lid; // we refer to the servo motor as 'lid'
void setup(){
 pinMode(trigPin, OUTPUT);// pin 7 is for sending information
 pinMode(echoPin, INPUT);// pin 6 is for receiving information
 lid.attach(9); // connect the servo motor to D9
}
void loop(){
 long duration, distance; // declare two variables to store time (duration) and distance
 digitalWrite(trigPin, LOW);// to start, make sure it's not sending an ultrasonic pulse
 delayMicroseconds(2);// wait for 2 microseconds to avoid interference
 digitalWrite(trigPin, HIGH);// send a pulse
 delayMicroseconds(10);// the pulse lasts for 10 microseconds
 digitalWrite(trigPin, LOW);// stop the pulse
 duration = pulseIn(echoPin, HIGH);// listen for the echo of the pulse to determine the duration
 distance = (duration/2) / 29.1;
 // The speed of sound is 340 meters per second, or equivalently, one centimeter every 29.1 microsecor
 // The ping travels from the sensor, bounces off the hand, and returns to the sensor.
 // The distance it covers is twice the actual distance. Therefore, "duration" needs to be divided by 2,
 // and the distance is the time divided by two and then divided by 29.1.
 if (distance < 10)
{// if the distance is less than 10 cm
   lid.write(90);// set the servo motor to ninety degrees (open mouth)
   delay(100);// wait for 100 milliseconds, i.e., 1/10 of a second
   lid.write(40);// set the servo motor to forty degrees (semi-open mouth)
   delay(100);// wait for 100 milliseconds, i.e., 1/10 of a second
   lid.write(90);// set the servo motor to ninety degrees (open mouth)
   delay(100);// wait for 100 milliseconds, i.e., 1/10 of a second
   lid.write(40);// set the servo motor to forty degrees (semi-open mouth)
   delay(100);// wait for 100 milliseconds, i.e., 1/10 of a second
   lid.write(90);// set the servo motor to ninety degrees (open mouth)
   delay(100);// wait for 100 milliseconds, i.e., 1/10 of a second
   lid.write(0);// set the servo motor to zero degrees (closed mouth)
 }
 else{ // if not,
   lid.write(0);// leave the servo motor at zero degrees (closed mouth)
 }
 delay(500);// wait for half a second before sending another pulse
}
```

Procedure:

1. Open tinker cad
2. Setup the circuit as per circuit diagram
3. Connect the components as shown in the connection diagram.
4. Connect power supply
5. Verify the output .

Result: